Statistics – Scenario-Based Questions (Set 4)

Scenario 1

Question: A stock brokerage firm wants to analyze historical stock prices to detect trends. How can Data Science concepts be applied?

Answer:

• I would first arrange the historical prices by date and check for missing values or duplicate records. This helps make sure the trend is based on reliable data.

• Next, I would plot the closing prices on a line chart and calculate daily price changes. A 7-day or 30-day moving average can make the overall direction easier to see.

• Finally, I would look at the highest and lowest prices, average returns and volatility. This would help the firm understand past price movements, but it would not guarantee future prices.

Scenario 2

Question: An educational institution has student score data. How can Data Science concepts be used to summarize the data effectively?

Answer:

• I would start by checking whether any student scores are missing or entered incorrectly. Then I would calculate the mean and median to understand the typical score.

• I would also check the minimum, maximum, quartiles and standard deviation to see how much the scores differ between students.

• A histogram would show how the marks are distributed, while a box plot would help identify unusually low or high scores. This makes it easier to see where students may need extra support.

Scenario 3

Question: An insurance company wants to analyze the distribution of claim amounts. How can Data Science concepts be applied?

Answer:

• First, I would check the claim amounts for missing entries, duplicate claims and unusual values. I would verify large claims before treating them as errors.

• Then I would calculate the mean, median, minimum, maximum and standard deviation. I would also check skewness, because a few expensive claims can increase the average.

• A histogram would show which claim amounts occur most often, and a box plot would highlight unusually large claims. The findings could help the company plan its claim reserves and review workload.

Scenario 4

Question: A healthcare organization wants to understand the relationship between patients' age, BMI, and blood pressure levels. How can Data Science concepts be applied?

Answer:

• I would check the age, BMI and blood pressure columns for missing or incorrect values before starting the analysis.

• Next, I would use correlation to understand whether these measurements tend to increase or decrease together. Scatter plots would help me see the patterns, and a heatmap would summarize the correlations.

• I would also review unusual observations instead of removing them automatically. A relationship in the data does not prove that one factor causes another.

Scenario 5

Question: A retail company has a customer database with incomplete and inconsistent data. How can Data Science concepts be used to clean and preprocess this data?

Answer:

• I would first remove duplicate customer records and check for missing information such as age, contact details or purchase history.

• I would standardize inconsistent entries, such as different spellings of the same city or mixed date formats. Missing values could be filled using a suitable method or left as unknown, depending on the column.

• Finally, I would check data types, identify suspicious values and validate the cleaned data. This would make later customer analysis more reliable.